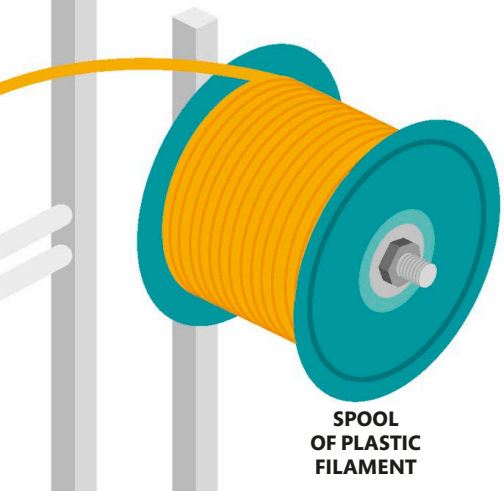
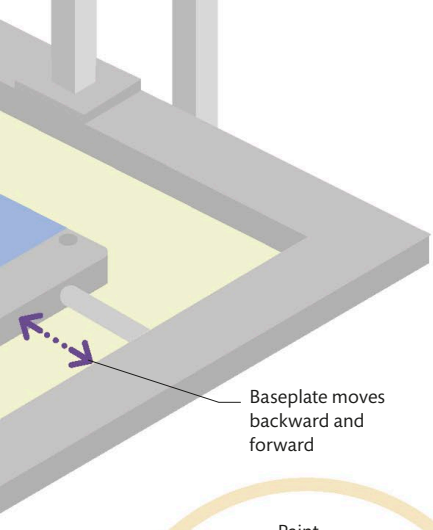
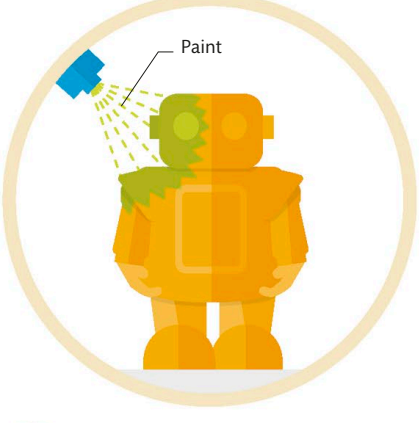



**4 Finishing**
Because of the layer-by-layer nature of the printing process, 3-D printed objects have a rough surface. It is usually necessary to treat them with chemicals or to smooth them mechanically to achieve a clean finish. They may also be painted.

## Uses of 3-D printers

3-D printing is still a young technology and is not yet commonly used to manufacture mass-produced consumer items. It is mainly utilized to produce specialized or custom-made items, such as pills and prosthetic body parts in medicine, musical instruments, and prototypes of potential new products.



**Pills**
3-D printing allows pharmaceutical manufacturers to better fine-tune the composition of pills compared with traditional pill-making methods. It also makes possible production of pills that dissolve almost instantly.



**Synthetic blood vessels**
Scientists have 3-D-printed blood vessels incorporating living cells. These vessels have been successfully implanted into mice and could in the future be used to replace damaged blood vessels in humans.



**Sports shoes**
Several sportswear companies have produced 3-D-printed sports shoes. They have been worn by athletes competing in international events but are available only in limited numbers.



**Prosthetic bones**
Some patients who have had a section of bone removed (to treat cancer, for example) have received 3-D-printed implants made of titanium or synthetic bone that are an exact match for the area of bone removed.



**Prosthetic limbs**
The use of 3-D printing to make prosthetic limbs has led to more lightweight designs than conventional prostheses. 3-D-printed limbs are also cheaper to produce and easier to customize for each individual.



**Musical instruments**
A wide variety of musical instruments have been experimentally 3-D printed, and many are available commercially, including some wind and string instruments, such as flutes, guitars, and violins.

## MADE IN SPACE

In 2014, astronauts on the International Space Station printed a ratchet wrench with a design file transmitted from the ground. 3-D printing could avoid the need to carry items that might never be used or to supply spare parts over large distances at great expense.

